

ALESSANDRA GUERINONI

📍 Palo Alto, CA 📞 415-307-5619 ✉ ale.guerinoni@gmail.com 🌐 alessandraguerinoni 📧 alegueri 🌐 alessandraguerinoni.com

SUMMARY

Senior Software Engineer and Tech Lead at Tesla with 4 years' experience designing distributed systems (Kafka, Kubernetes, Go) and shipping AI-powered tooling at scale. Two promotions in 3 years; portfolio of \$22M+/year in validated business impact.

EDUCATION

Bachelor of Engineering with Distinction, Mechatronics Engineering

Sept 2016 - Apr 2022

McMaster University

- Awards: Dean's Honour List (multi-time); Golden Key Award (top 15% of the faculty); Future Leader Recognition Award; Faculty of Engineering Entrance Award.

Machine Learning Certificate

May 2024 - Aug 2024

Cornell University

WORK EXPERIENCE

Senior Software Engineer, Tech Lead

July 2022 - Present

Tesla

- Advanced from Associate to Senior Software Engineer in under 3 years (two promotions, Apr 2025).
- Recruited to a critical org-wide engineering platform migration (estimated \$100M in projected savings once complete); validated APIs, coordinated data migration, and unblocked change management for design data across the company.
- Built a Kafka/Kubernetes/Go notification pipeline that gave multiple teams real-time access to engineering data from the platform, replacing manual polling and unlocking downstream automation; \$18.8M in validated business impact.
- Architected and shipped a full-stack web application (Next.js, React, Go microservices, Kafka, S3) replacing legacy tooling — 10X performance improvement, 99.76% uptime; reduced approval cycle time from ~20 days to 8 days (60% reduction); \$3.1M/year validated savings.
- Spearheaded AI integration into engineering workflows — production features achieving 97% classification accuracy, eliminating ~6,000 hours/year of manual review (\$1.2M/year validated savings).
- Built ETL pipelines (Go, MySQL, Temporal) exposing engineering data to internal services via gRPC and REST; ~\$10M in validated business impact.
- Mentored engineers and delegated major features while maintaining technical accountability; contributed to the hiring pipeline through technical screens and panel interviews; mentored intern through to full-time conversion.
- Assumed Acting Product Manager responsibilities (Apr 2026 - Present), owning backlog prioritization, product strategy, and stakeholder alignment after PM departure.

Project Manager, Governance & University Collaboration

May 2019 - Aug 2020

Bombardier, Mitsubishi Heavy Industries

- Managed 50+ enterprise projects; led a team of 6 coordinating university partnerships and recruiting 26 interns; negotiated contracts reducing cost overhead by 2%.
- Built Python tooling for Oracle database access to streamline project reporting.

Teaching Assistant – Discrete Mathematics & Computational Thinking

Jan 2019 - Apr 2019 & Sept 2021 - Dec 2021

McMaster Faculty of Engineering

- Facilitated Haskell and computer science learning for 50+ students; led tutorials, graded assignments, and ran consultations.

Software Engineering Camp Counselor

Apr 2018 - Sept 2018

McMaster Faculty of Engineering

- Introduced 250+ children to Software Engineering through Python, Arduino, and web projects; delivered STEM lectures to 100+ students across Hamilton.

PROJECTS

TriAuto — Real-Time Decision-Support System

Sept 2021 - Apr 2022

McMaster University Capstone

- Co-developed with a 4-person team a full-stack web application that ingests structured input data, applies a classification ruleset, and pushes recommendations to a live dashboard over WebSocket. Stack: React (client), Node.js + Socket.IO (real-time backend), MySQL (persistence).

SKILLS

Languages:

Go • TypeScript • JavaScript • Python • Java • C++ • SQL

Backend & Data:

Kafka • Temporal • MySQL • Redis • gRPC • Protobuf • REST • OAuth • JWT

Cloud & DevOps:

AWS • Azure • Docker • Kubernetes • S3 • Jenkins • ArgoCD • GitHub Actions • Prometheus • Grafana • Splunk

Frontend:

Next.js • React • Node.js